Q.① Branch point & Branch cut of $\ln(z^2-1)$.

Branch point $\rightarrow$ $z^2-1=0$

$\Rightarrow \boxed{z=\pm 1}$

Branch cut $\rightarrow$ $Re(z^2-1) \leq 0$

& $Im(z^2-1)=0$

i.e. $Re(z^2-1) \leq 0$

$Re.(x^2-y^2+2ixy-1) \leq 0$

i.e. $x^2-y^2-1 \leq 0$

$Im(z^2-1)=0$

$\Rightarrow 2xy=0$

$\Rightarrow x=0$ or $y=0$

If $x=0$ then $x^2-y^2-1 \leq 0$ implies that

$y^2+1 \geq 0$

i.e. $x=0$, whole y-axis.

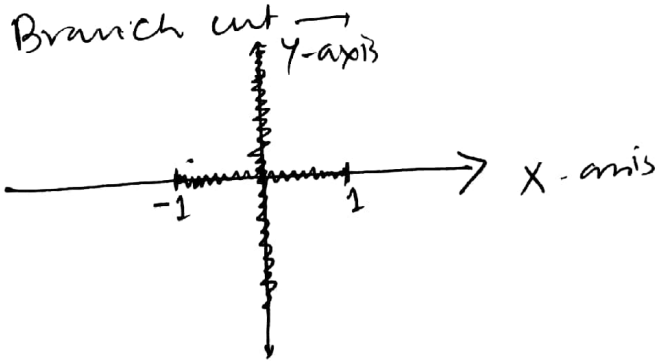
If $y=0$ then $x^2-y^2-1 \leq 0$ implies that

$x^2-1 \leq 0$

i.e. $x^2 \leq 1$

i.e. $x \in [-1,1]$.

Therefore Branch cut $\rightarrow$



we have to write a branch of $\ln(z)$ in $\{(r,\theta) \mid r>0,\ \theta \in (\pi, 3\pi)\}$.

Ans:- $\ln(z) = \ln|r| + i \cdot \underbrace{Arg_{\pi}(z)}$

$Arg_{\pi}(z) \in (\pi, 3\pi)$ # Ans

Q-(3) $\int_C z^n \, dz = \begin{cases} 2\pi i & \text{if } n=-1 \\ 0 & \text{if } n \neq -1 \end{cases}$

where C is $|z| = R$

$\Rightarrow$ Sol^n:- See Example 11.7 in Iyengar & Jain Book.

Q.5 $\longrightarrow$ Sol^n mentioned in Iyengar & Jain Book (see Example 11.9)

Q.6 (i) $\int_{|z|=R} (\bar{z})^n \, dz \longrightarrow$ see the lecture notes

(ii) $\int_{|z|=1} \sqrt{z} \, dz \longrightarrow$ See the lecture notes

(iii) $\int_{|z|=1} \frac{e^{5z}}{z^3} \, dz \longrightarrow$ See the lecture notes.

(iv) $\int_{|z|=9} \sin z \, dz = 0$ Since $\sin z$ is analytic for throughout the complex plane.

Hence By cauchy theorem.

$\int_{|z|=9} \sin z \, dz = 0$ #

Q.7 (i) find out upper bound of

$$\left| \int_{|z|=2} e^{\bar{z}^2} \, dz \right|$$

$\Rightarrow$ see the lecture notes.

(ii) find out M s.t. $\left| \int_C \frac{e^{iz}}{(1+z^2)} dz \right| \leq M$ with

$C: \gamma(t) = Re^{it}$, $\underline{t \in [0, \pi]}$, $\underline{R > 1}$

Now $|1+z^2| \geq |1-|z|^2| = |1-R^2| = \underline{(R^2-1)}$

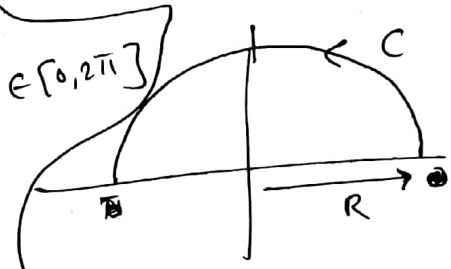
(since $R>1$)

Now on C, we have

$$|f(z)| = \left| \frac{e^{iz}}{(1+z^2)} \right| \leq \frac{|e^{iz}|}{(R^2-1)} = \frac{|e^{ix} \cdot e^{-y}|}{(R^2-1)}$$

$$= \frac{|e^{-y}|}{(R^2-1)} \leq \frac{1}{(R^2-1)}$$

Note
But if we take
$C: \gamma(t) = Re^{it}$, $t \in [0, 2\pi]$
then
$$|f(z)| \leq \frac{|e^{-y}|}{(R^2-1)} \leq \frac{e^R}{(R^2-1)}$$



Hence we have

$$\left| \int_C \frac{e^{iz}}{(1+z^2)} dz \right| \leq \frac{1}{(R^2-1)} \cdot (\pi \times R)$$

$$= \frac{\pi R}{(R^2-1)}$$

But for

$$\left| \int_{|z|=R} \frac{e^{iz}}{(1+z^2)} dz \right| \leq \frac{e^R \cdot (2\pi R)}{(R^2-1)} = \frac{2\pi R e^R}{(R^2-1)}$$

Q.(8) Find out $\int_C z e^{z^2} dz$ where,

(i) $C \rightarrow$ a line segment from i to $-i+2$

(ii) $C \rightarrow$ a curve from 0 to $(1, 1)$ along $y = x^2$.

$\Rightarrow$ $g(z) = \frac{1}{2} e^{z^2}$ is primitive of the fn $f(z) = z e^{z^2}$. Therefore $\rightarrow$

(i) $\int_C z e^{z^2} dz = g(-i+2) - g(i)$

$= \frac{1}{2}(e^{3-4i} - e^{-1})$ # Ans

(ii) $\int_C z e^{z^2} dz = g(1+i) - g(0) = \frac{1}{2}(e^{2i} - 1)$

Ans. #

Q(9) Evaluate the integral over unit circle $\rightarrow$

(i) $\int_{|z|=1} \frac{\cos z}{z} dz = 2\pi i \cdot \cos(0) = 2\pi i$

(ii) $\int_{|z|=1} \frac{\sin^2 z}{z} dz = 2\pi i \cdot \sin^2 0 = 0$

(iii) $\int_{|z|=1} \frac{\cos z^2}{z} dz = 2\pi i \cdot \cos 0 = 2\pi i$

(iv) $\int_{|z|=1} \frac{e^z \cos z}{(z - \pi/4)^3} dz = \frac{2\pi i}{2} f^{(2)}(\pi/4)$ — (1)

where $f(z) = e^z \cos z$

find out $f''(z)$ and then put the value $f''(\pi/4)$ in eq^n (1).

Ans

Q. 10 $\int_C (z-i)\,dz$, where $C := \gamma(t) = (t, t^2)$

$z(t) = t + t^2 i \quad t \in [-1, 1]$

$z'(t) = (1 + 2ti)$

$$= \int_{-1}^{1} \left[(t + t^2 i) - i \right] \cdot (1 + 2ti)\, dt$$

$$= \int_{-1}^{1} (t + (t^2 - 1) i) \cdot (1 + 2ti)\, dt$$

$$= \int_{-1}^{1} \left[t + 2t^2 i + (t^2 - 1) i - 2t(t^2 - 1) \right] dt$$

$$= \int_{-1}^{1} \left[3t - 2t^3 + (3t^2 - 1) i \right] dt$$

$$= \left[3\frac{t^2}{2} - \frac{2t^4}{4} \right]_{-1}^{1} + i \left[3\frac{t^3}{3} - t \right]_{-1}^{1}$$

$$= \left(\frac{3}{2} - \frac{1}{2} \right) - \left(\frac{3}{2} - \frac{1}{2} \right) + i \left[(1 - 1) - (-1 + 1) \right]$$

$$= 0$$

Ans.

Q. 11 (i) $\int_{|z|=1} \frac{e^{2z}}{z^4}\, dz \longrightarrow$ (See Churchill Brown book page 162.)

(ii) $\int_{|z-i|=2} \frac{1}{(z^2+4)^2}\, dz = \int_{|z-i|=2} \frac{1}{(z+2i)^2 (z-2i)^2}\, dz$

$= \int_{|z-i|=2} \frac{f(z)}{(z-2i)^2}\, dz$ & $f(z) = \frac{1}{(z+2i)^2}$

$\Downarrow$

$= \frac{2\pi i}{1!} f'(2i)$

(iii) $\int_{|z|=1} \sec z\, dz = 0$ (see Kreyszig page 653)

analytic on $|z|=1$ & inside $|z|=1$

$\Rightarrow$ By Cauchy theorem we get zero value of the integral.

Ans.

Q.(12) $\int_C \frac{\sin z}{(z+4iz)}\,dz$, $C :\Rightarrow |z-4-2i| = 6.5$

or $|z-(4+2i)| = 6.5$

$$\int_C \frac{\sin z}{z(1+4i)}\,dz = \int_C \frac{\sin z/(1+4i)}{z}\,dz$$

$$= \frac{1}{(1+4i)}\cdot(2\pi i \cdot \sin 0) = 0$$

Ans

Q.(13) $\int_C \frac{\text{Ln}(z+1)}{(z^2+1)}\,dz$ $\quad |z-i| = 1.4.$

$$= \int_C \frac{\text{Ln}(z+1)}{(z+i)(z-i)}\,dz$$

$$= 2\pi i\, g(i) \quad \text{where} \quad \boxed{g(z) = \frac{\text{Ln}(z+1)}{(z+i)}}$$

Ans.

Q.(14) $\int_C \bar{z}\,dz \rightarrow$ See the lecture notes.

Q(15) (i) $\int_0^i (z^2+1)^3\,dz$

Take the straight line between 0 to i

$z(t) = 0 + ti \quad 0 \le t \le 1$

$z'(t) = i$

$$\int_0^1 \left((ti)^2+1\right)^3 \cdot (i)\,dt = \int_0^1 (-t^2+1)^3 \cdot (i)\,dt$$

just solve the integral & find the value.

(ii) $\displaystyle\int_0^1 \frac{\tan^{-1} z}{(1+z^2)}\, dz$

$f(z) = \dfrac{\tan^{-1} z}{1+z^2}$ is analytic in the rectangle doesn't contain i or $-i$

take path $z(t) = t + 0 \cdot i$ $\quad 0 \le t \le 1$

$(z'(t) = 1)$

$$\int_0^1 \frac{\tan^{-1} t}{(1+t^2)} \cdot 1\, dt$$

$$= \pi^2/32 \quad \text{Ans.}$$

(16) (i) $\displaystyle\int_C \frac{dz}{(z-1)(z+i)}$, where C contains 1 & $-i$

$$= \int_C \frac{1}{(1+i)} \left[\frac{1}{(z-1)} - \frac{1}{(z+i)} \right]$$

$$= \frac{1}{(1+i)} \left[\int_C \frac{1}{(z-1)}\, dz - \int_C \frac{1}{(z+i)}\, dz \right]$$

$$= \frac{1}{(1+i)} \cdot \left[2\pi i - 2\pi i \right] = 0$$

(ii) $\displaystyle\int_{|z|=1} \frac{dz}{(z^3 + 3iz^2)} = \int_{|z|=1} \frac{dz}{z^2 (z+3i)}$

$$= \int_{|z|=1} \frac{\left(\frac{1}{z+3i}\right) dz}{z^2}$$

$$= 2\pi i\, g^{(1)}(0), \text{ where } g(z) = \frac{1}{(z+3i)}$$

Ans.